

SYLLABUS

COURSE NAME : DATA BASE MANAGEMENT SYSTEM

COURSE CODE : CSE335

COURSE TYPE : Theory

OFFERED PROGRAMMES:

Bachelor of Technology (Artificial Intelligence and Machine Learning) (SEMESTER - III), Bachelor of Technology (Computer Science and Engineering - Cyber Security) (SEMESTER - III), Bachelor of Technology (Computer Science and Engineering) (SEMESTER - III), Bachelor of Technology (Computer Science and Engineering-Artificial Intelligence and Machine Learning) (SEMESTER - III), Bachelor of Technology (Computer Science and Engineering-Data Science) (SEMESTER - III), Bachelor of Technology (Computer Science and Engineering-IoT) (SEMESTER - III)

Course learning Outcomes

Outcome Reference No.	Outcome Description	RBT Domain	Highest RBT Level
CO1	Apply the Concepts of Entity-Relationship (E-R) model for the given application	Cognitive	L3 - Apply
CO2	Apply Normalization principles to create and manipulate relational databases	Cognitive	L3 - Apply
CO3	Examine different file organization concepts for data storage in Relational databases	Cognitive	L4 - Analyse
CO4	Utilize the transaction management principles on relational databases	Cognitive	L3 - Apply
CO5	Apply the concepts of Non-Relational Models	Cognitive	L3 - Apply

Sustainability Development Goals (SDG)

- Industry, innovation and infrastructure

21st Century Skills

- Technological literacy

Course Credit Details

Display Name	Course Credit	Hours Per Session
Theory	3	45

Course Content

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U1	Introduction and Conceptual Modeling	9	Local	Skill Development	None	-

Topic No.	Content	Description	Topic Reference Material Upload
U1-T1	Introduction to File and Database systems- Database system structure – Data Models – Introduction to Network and Hierarchical Models – ER model – Relational Model – Relational Algebra.	-	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U2	Relational Model	9	Regional	Employability	None	-

Topic No.	Content	Description	Topic Reference Material Upload
U2-T1	SQL – Data definition- Queries in SQL- Updates- Views – Integrity and Security – Relational Database design – Functional dependencies and Normalization for Relational Databases (up to BCNF).	-	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U3	Storage Management & Query Processing	9	Regional	None	None	-

Topic No.	Content	Description	Topic Reference Material Upload
U3-T1	Record storage and Primary file organization- Secondary storage Devices- Operations on Files- Heap File- Sorted Files-	-	-

Topic No.	Content	Description	Topic Reference Material Upload
	Hashing Techniques – Index Structure for files –Different types of Indexes- B-Tree - B+ Tree – Query Processing.		

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U4	Transaction Management	9	National	None	None	-

Topic No.	Content	Description	Topic Reference Material Upload
U4-T1	Transaction management -Transaction Processing – Introduction- Need for Concurrency control- Desirable properties of Transaction- Schedule and Recoverability- Serializability and Schedules – Concurrency Control – Types of Locks- Two Phases locking- Deadlock- Time stamp based concurrency control – Recovery Techniques – Concepts- Immediate Update- Deferred Update - Shadow Paging.	-	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U5	Current Trends	9	Global	None	None	-

Topic No.	Content	Description	Topic Reference Material Upload
U5-T1	Introduction to NOSQL Systems, The CAP Theorem, Document-Based NOSQL Systems and MongoDB, NOSQL Key-Value Stores, Column-Based or Wide Column NOSQL Systems, NOSQL Graph Databases and Neo4j.	-	-

Text & Reference Books

T1. Abraham Silberschatz, Henry F. Korth and S. Sudarshan- “Database System Concepts”, Seventh Edition, McGraw-Hill, 2021.

T1. Andreas Meier · Michael Kaufmann "SQL & NoSQL Databases", Springer -2019.

R1. Raghu Ramakrishnan, “Database Management System”, Tata McGraw-Hill Publishing Company, 2003

EXAMINATION PATTERN

PATTERN FOR CONTINUOUS INTERNAL ASSESSMENT (CIA)

CIA-1-20 Marks- Two Components

CIA-2-50 Marks (Centralised)

CIA-3-20 Marks Two Components OR One Mini Project

QUESTION PAPER PATTERN FOR END SEMESTER EXAMINATION (ESE)

5 Questions with Either or Choice for all five units. Each carries 20 Marks.

Course Plan

COURSE NAME : DATA BASE MANAGEMENT SYSTEM

COURSE CODE : CSE335

SESSION GROUP : ODD

DEPARTMENT : COMPUTER SCIENCE AND ENGINEERING

Course Type	Credits	Total Hours
Theory	3	45

Sustainability Development Goals (SDG)

- Industry, innovation and infrastructure

21st Century Skills

- Technological literacy

DESCRIPTION

This course introduces the fundamentals of Database Management Systems (DBMS), including database design, data models, and normalization. Students learn to create and manage relational databases using SQL and tools like MySQL, and develop skills in data querying, analysis, and management, preparing them for roles in data-driven industries.

COURSE OBJECTIVES

1. Understand Fundamental DBMS Concepts: Students will learn the fundamental concepts of database management systems, including data models, data normalization, and database design. 2. Design and Implement Relational Databases: Students will design and implement relational databases using SQL and manage them using popular DBMS tools like MySQL. 3. Query and Manipulate Data with SQL: Students will use SQL to query and manipulate data stored in databases.

Unit No	Unit Title	Unit Teaching Hours	Development Needs	Skills Focused	Cross Cutting Issues
U1	Introduction and Conceptual Modeling	9	Local	Skill Development	None
U2	Relational Model	9	Regional	Employability	None
U3	Storage Management & Query Processing	9	Regional	None	None
U4	Transaction Management	9	National	None	None
U5	Current Trends	9	Global	None	None

Lesson Plan

Week	Unit	Topics	Andragogy	Ungraded Formative Assessments	Reference Resources	Additional Details
Week 1	U1	U1-T1	Problem based learning	Quiz - ICT Tools	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan- "Database System Concepts", Seventh Edition, McGraw-Hill, 2021	
Week 2	U1	U1-T1	Experientia l learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan- "Database System Concepts", Seventh Edition, McGraw-Hill, 2021	
Week 3	U1	U1-T1	Experientia l learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan- "Database System Concepts", Seventh Edition, McGraw-Hill, 2021	
Week 4	U2	U2-T1	Experientia l learning	NIL	T1.: Abraham Silberschatz,	

Week	Unit	Topics	Andragogy	Ungraded Formative Assessments	Reference Resources	Additional Details
					Henry F. Korth and S. Sudarshan- “Database System Concepts”, Seventh Edition, McGraw-Hill, 2021	
Week 5	U2	U2-T1	Experiential learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan- “Database System Concepts”, Seventh Edition, McGraw-Hill, 2021	
Week 6	U2	U2-T1	Experiential learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan- “Database System Concepts”, Seventh Edition, McGraw-Hill, 2021	
Week 7	U3	U3-T1	Experiential learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan- “Database System	

Week	Unit	Topics	Andragogy	Ungraded Formative Assessments	Reference Resources	Additional Details
					Concepts”, Seventh Edition, McGraw-Hill, 2021	
Week 8	U3	U3-T1	Experiential learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan-“Database System Concepts”, Seventh Edition, McGraw-Hill, 2021	
Week 9	U3	U3-T1	Experiential learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan-“Database System Concepts”, Seventh Edition, McGraw-Hill, 2021	
Week 10	U4	U4-T1	Experiential learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan-“Database System Concepts”, Seventh Edition, McGraw-Hill, 2021	
Week 11	U4	U4-T1	Experiential learning	NIL	T1.: Abraham	

Week	Unit	Topics	Andragogy	Ungraded Formative Assessments	Reference Resources	Additional Details
			I learning		Silberschatz, Henry F. Korth and S. Sudarshan- "Database System Concepts", Seventh Edition, McGraw-Hill, 2021	
Week 12	U4	U4-T1	Experientia I learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan- "Database System Concepts", Seventh Edition, McGraw-Hill, 2021	
Week 13	U5	U5-T1	Experientia I learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan- "Database System Concepts", Seventh Edition, McGraw-Hill, 2021	
Week 14	U5	U5-T1	Experientia I learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan- "Database	

Week	Unit	Topics	Andragogy	Ungraded Formative Assessments	Reference Resources	Additional Details
					System Concepts", Seventh Edition, McGraw-Hill, 2021	
Week 15	U5	U5-T1	Experiential learning	NIL	T1.: Abraham Silberschatz, Henry F. Korth and S. Sudarshan- "Database System Concepts", Seventh Edition, McGraw-Hill, 2021	

CO-PO Mapping

Programme: Bachelor of Technology (Computer Science and Engineering - Cyber Security)

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	0	1	2	-	-	-	-	-	-	-
CO2	3	2	0	1	1	-	-	-	-	-	-	-
CO3	3	3	1	2	1	-	-	-	-	-	-	-
CO4	3	2	0	1	1	-	-	-	-	-	-	-
CO5	3	2	0	1	1	-	-	-	-	-	-	-

CO-PSO Mapping

CO/PSO	PSO1	PSO2	PSO3
CO1	0	0	1
CO2	1	2	1
CO3	1	1	0
CO4	0	1	0
CO5	0	1	0

Programme: Bachelor of Technology (Computer Science and Engineering-IoT)

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	0	1	2	-	-	-	-	-	-	-
CO2	3	2	0	1	1	-	-	-	-	-	-	-
CO3	3	3	1	2	1	-	-	-	-	-	-	-
CO4	3	2	0	1	1	-	-	-	-	-	-	-
CO5	3	2	0	1	1	-	-	-	-	-	-	-

CO-PSO Mapping

CO/PSO	PSO1	PSO2	PSO3
CO1	0	0	1
CO2	1	2	1
CO3	1	1	0
CO4	0	1	0
CO5	0	1	0

Programme: Bachelor of Technology (Computer Science and Engineering-Data Science)

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	0	1	2	-	-	-	-	-	-	-
CO2	3	2	0	1	1	-	-	-	-	-	-	-
CO3	3	3	1	2	1	-	-	-	-	-	-	-
CO4	3	2	0	1	1	-	-	-	-	-	-	-
CO5	3	2	0	1	1	-	-	-	-	-	-	-

CO-PSO Mapping

CO/PSO	PSO1	PSO2	PSO3
CO1	0	0	1
CO2	1	2	1
CO3	1	1	0
CO4	0	1	0
CO5	0	1	0

Programme: Bachelor of Technology (Computer Science and Engineering-Artificial Intelligence and Machine Learning)

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	0	1	2	-	-	-	-	-	-	-
CO2	3	2	0	1	1	-	-	-	-	-	-	-
CO3	3	3	1	2	1	-	-	-	-	-	-	-
CO4	3	2	0	1	1	-	-	-	-	-	-	-
CO5	3	2	0	1	1	-	-	-	-	-	-	-

CO-PSO Mapping

CO/PSO	PSO1	PSO2	PSO3
CO1	0	0	1
CO2	1	2	1
CO3	1	1	0
CO4	0	1	0
CO5	0	1	0

Programme: Bachelor of Technology (Computer Science and Engineering)

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	0	1	2	-	-	-	-	-	-	-
CO2	3	2	0	1	1	-	-	-	-	-	-	-
CO3	3	3	1	2	1	-	-	-	-	-	-	-
CO4	3	2	0	1	1	-	-	-	-	-	-	-
CO5	3	2	0	1	1	-	-	-	-	-	-	-

CO-PSO Mapping

CO/PSO	PSO1	PSO2	PSO3
CO1	0	0	1
CO2	1	2	1
CO3	1	1	0
CO4	0	1	0
CO5	0	1	0

Programme: Bachelor of Technology (Artificial Intelligence and Machine Learning)

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	0	1	2	-	-	-	-	-	-	-

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	2	0	1	1	-	-	-	-	-	-	-
CO3	3	3	1	2	1	-	-	-	-	-	-	-
CO4	3	2	0	1	1	-	-	-	-	-	-	-
CO5	3	2	0	1	1	-	-	-	-	-	-	-

CO-PSO Mapping

CO/PSO	PSO1	PSO2	PSO3
CO1	0	0	1
CO2	1	2	1
CO3	1	1	0
CO4	0	1	0
CO5	0	1	0

CO Assessment Mapping

Assessment Rule Setting for Theory

Assessment	Assessment Levels	Total	CO1	CO2	CO3	CO4	CO5
CIA - I	Department Level	20.00	10	10	0	0	0
CIA - II	Centralised Level	50.00	20	20	10	0	0
CIA - III	Department Level	20.00	0	0	0	10	10
End Term	Centralised Level	100.00	20	20	20	20	20

Assessment Component Definition

Assessment Component Definition Type: COMMON

Course Type: Theory

Assessment: CIA - I - Total Marks: 20

Total Marks	Scale Down To	Min Marks
20	-	-

CO Allocation	
CO1	CO2
10	10

Component	Description/Rubrics
Case Analysis	Description:

Component	Description/Rubrics
	CIA 1 – Component 1 ER MODEL DESIGN: I. Learning Objectives: To understand the ER Model and to design an ER for any enterprise To convert the ER model into a Relational Model To utilize the library resources Marks: 10 Marks Tentative Date: Announced in Class Venue: offline Pre-Assessment Criteria: Proper Design considering all the aspects and symbols of ER and a good presentation: - (HIGH) Average Design (MEDIUM) Poor Design and presentation (LOW) Learning Objective To enable students to understand well the data modeling concepts and to relate to a practical scenario To test the presentation skills in communication, writing, and drawing Assignment Learning Objectives To enable students to understand the data modeling concepts well and to relate to a practical scenario

Component	Description/Rubrics																													
	To test the presentation skills in communication, writing, and drawing Assessment Strategies aligned to LO: ER Diagram Design with proper analysis, Proper organization, and logical flow. Technology Tools used along with their Purpose: LM ER DIAGRAM - Students will select a topic group wise on different Data base models Students analyze a real-world problem/domain and design ER models based on requirements. Rubrics: Evaluation Rubrics: <table><tr><th>PARAMETERS / SCORE</th><th>10</th><th>7</th><th>5</th><th>3</th></tr><tr><td>Problem statement</td><td>A Complete response with detailed explanation</td><td>Good response with clear information</td><td>Explanation not well defined</td><td>Poor Explanation/ Problem identification</td></tr><tr><td>Use of Visuals</td><td>Clear diagram with details</td><td>Clear diagram with incomplete details</td><td>In-appropriate diagram</td><td>No diagram</td></tr><tr><td>Knowledge Demonstration</td><td>Shows complete process of ER Diagram to Schema mapping</td><td>Shows substantial process of ER Diagram to Schema mapping</td><td>Weak implementation of ER Diagram to Schema mapping</td><td>Lack of understanding</td></tr><tr><td>Usage of Library resources</td><td>Proper citations mentioned in the poster</td><td>Citations are partially relevant</td><td>Citations are not much relevant</td><td>Citations are not available.</td></tr></table>					PARAMETERS / SCORE	10	7	5	3	Problem statement	A Complete response with detailed explanation	Good response with clear information	Explanation not well defined	Poor Explanation/ Problem identification	Use of Visuals	Clear diagram with details	Clear diagram with incomplete details	In-appropriate diagram	No diagram	Knowledge Demonstration	Shows complete process of ER Diagram to Schema mapping	Shows substantial process of ER Diagram to Schema mapping	Weak implementation of ER Diagram to Schema mapping	Lack of understanding	Usage of Library resources	Proper citations mentioned in the poster	Citations are partially relevant	Citations are not much relevant	Citations are not available.
PARAMETERS / SCORE	10	7	5	3																										
Problem statement	A Complete response with detailed explanation	Good response with clear information	Explanation not well defined	Poor Explanation/ Problem identification																										
Use of Visuals	Clear diagram with details	Clear diagram with incomplete details	In-appropriate diagram	No diagram																										
Knowledge Demonstration	Shows complete process of ER Diagram to Schema mapping	Shows substantial process of ER Diagram to Schema mapping	Weak implementation of ER Diagram to Schema mapping	Lack of understanding																										
Usage of Library resources	Proper citations mentioned in the poster	Citations are partially relevant	Citations are not much relevant	Citations are not available.																										

Component	Description/Rubrics																		
Scenario-based analysis	Description: CIA 1 – Component 2 Assignment Description Assignment Type : Scenario Based Question Marks : 10 Time : 1 hour Tentative Date : The exact date and deadline will be announced in the classroom. Syllabus: Unit 2 Remedial Class: Tentative Dates to be given with no. of classes to be conducted Learning Outcome(s): Students will learn about the basics of SQL commands Post–Assessment: Identify students scoring below 60% of the total marks and guide them with remedial classes. Students will be provided with different types of scenario based question. Rubrics: Evaluation Rubrics: <table> <tr> <th>Parameters / Score</th><th>10</th><th>7</th><th>5</th><th>2</th></tr> <tr> <td>Problem Understanding</td><td>Demonstrates excellent understanding of the scenario with accurate interpretation.</td><td>Shows good understanding with minor gaps in interpretation.</td><td>Understands the scenario partially with limited clarity.</td><td>Shows minimal understanding of the scenario.</td></tr> <tr> <td>Logical Analysis</td><td>Provides highly</td><td>Analysis is</td><td>Analysis is</td><td>Limited logical</td></tr> </table>				Parameters / Score	10	7	5	2	Problem Understanding	Demonstrates excellent understanding of the scenario with accurate interpretation.	Shows good understanding with minor gaps in interpretation.	Understands the scenario partially with limited clarity.	Shows minimal understanding of the scenario.	Logical Analysis	Provides highly	Analysis is	Analysis is	Limited logical
Parameters / Score	10	7	5	2															
Problem Understanding	Demonstrates excellent understanding of the scenario with accurate interpretation.	Shows good understanding with minor gaps in interpretation.	Understands the scenario partially with limited clarity.	Shows minimal understanding of the scenario.															
Logical Analysis	Provides highly	Analysis is	Analysis is	Limited logical															

Component	Description/Rubrics				
		logical and well-structured analysis with strong reasoning.	mostly logical with good reasoning.	moderate but lacks depth.	analysis with weak reasoning.
	Solution Design / Approach	Develops an accurate, innovative, and complete solution approach.	Develops a suitable solution with minor errors.	Provides a basic solution with incomplete approach.	Solution approach is weak and poorly structured.
	Presentation & Documentation	Information is presented clearly with excellent organization and formatting.	Well-presented with good organization.	Presentation is average with some formatting issues.	Poor presentation and unclear documentation.

Assessment: CIA - III - Total Marks: 20

Total Marks	Scale Down To	Min Marks
20	-	-

CO Allocation	
CO4	CO5
10	10

Component	Description/Rubrics
Project submission	Description: Component 1: Presentation & Demonstration Marks: 10 Duration: 30 Minutes (Demonstration) Date: The exact date and deadline will be announced in the classroom Assignment Learning Objectives:

Component	Description/Rubrics																							
	-To enable students to understand the development of the project and give an effective presentation for the problems. -To test the student Involvement in the project and Proper Viva on the inclusion of SQL and a Demo. Assessment Strategies aligned to LO: Mini Project Development and Implementation Evaluation and Project Presentation Assessment. Technology Tools used along with their Purpose: PPT and Databases. Rubrics: Evaluation Rubrics: Phase 1: Presentation & Demonstration (10 Marks) <table> <tr> <th>Parameters / Score</th><th>10-9</th><th>8-7</th><th>6-5</th><th>4-0</th></tr> <tr> <td>Problem Understanding & Concept Clarity</td><td>Clearly explains the project objectives, problem statement, and database concepts with strong understanding.</td><td>Explains the project well with minor gaps in clarity.</td><td>Basic understanding with limited explanation of concepts.</td><td>Poor understanding and unclear explanation of the project.</td></tr> <tr> <td>Presentation Skills</td><td>Excellent communication, confidence, organization, and professional presentation.</td><td>Good presentation with minor communication issues.</td><td>Average presentation with limited confidence and organization.</td><td>Poor presentation and ineffective communication skills.</td></tr> <tr> <td>Demonstration of Project</td><td>Demonstrates all functionalities effectively with proper workflow explanation.</td><td>Demonstrates most functionalities with minor issues.</td><td>Partial demonstration with limited functionality explanation.</td><td>Incomplete or unclear demonstration of the project.</td></tr> </table>				Parameters / Score	10-9	8-7	6-5	4-0	Problem Understanding & Concept Clarity	Clearly explains the project objectives, problem statement, and database concepts with strong understanding.	Explains the project well with minor gaps in clarity.	Basic understanding with limited explanation of concepts.	Poor understanding and unclear explanation of the project.	Presentation Skills	Excellent communication, confidence, organization, and professional presentation.	Good presentation with minor communication issues.	Average presentation with limited confidence and organization.	Poor presentation and ineffective communication skills.	Demonstration of Project	Demonstrates all functionalities effectively with proper workflow explanation.	Demonstrates most functionalities with minor issues.	Partial demonstration with limited functionality explanation.	Incomplete or unclear demonstration of the project.
Parameters / Score	10-9	8-7	6-5	4-0																				
Problem Understanding & Concept Clarity	Clearly explains the project objectives, problem statement, and database concepts with strong understanding.	Explains the project well with minor gaps in clarity.	Basic understanding with limited explanation of concepts.	Poor understanding and unclear explanation of the project.																				
Presentation Skills	Excellent communication, confidence, organization, and professional presentation.	Good presentation with minor communication issues.	Average presentation with limited confidence and organization.	Poor presentation and ineffective communication skills.																				
Demonstration of Project	Demonstrates all functionalities effectively with proper workflow explanation.	Demonstrates most functionalities with minor issues.	Partial demonstration with limited functionality explanation.	Incomplete or unclear demonstration of the project.																				

Component	Description/Rubrics				
	Response to Questions	Answers all technical questions accurately with strong domain knowledge.	Responds well with minor technical gaps.	Basic responses with limited technical understanding.	Unable to answer technical questions properly.
Project submission	Description: Component 2: Implementation, Front-End & Back-End Integration & Report Submission Marks: 10 Duration: 30 Minutes Date: The exact date and deadline will be announced in the classroom Assignment Learning Objectives: -To enable students to understand the development of the front end and backend connection and to execute the project for the identified problems. -To test the explanation skills in narrating the project work. -Quality of the report contents and submission Assessment Strategies aligned to LO: Apply database concepts to develop real-world applications using Oracle, MongoDB, and MySQL. Technology Tools used along with their Purpose: Oracle/MongDB/Mysql... Rubrics: Evaluation Rubrics: Phase 2: Implementation with Front-End & Back-End Integration & Report Submission (10 Marks)				

Component	Description/Rubrics				
	Parameters / Score	10-9	8-7	6-5	4-0
	Database Implementation	Efficient implementation of tables, relationships, normalization, and SQL queries.	Good implementation with minor errors.	Basic implementation with limited database functionalities.	Poor or incomplete database implementation.
	Front-End & Back-End Integration	Excellent integration with smooth connectivity and proper user interaction.	Good integration with minor functional issues.	Partial integration with limited connectivity.	Integration is incomplete or not functioning properly.
	Functionality & Testing	All modules function correctly with proper testing and validation.	Most functionalities work with minor issues.	Limited functionality and incomplete testing.	Major functional errors and insufficient testing.
	Report Documentation	Comprehensive, well-structured report with proper formatting and screenshots.	Good documentation with minor formatting issues.	Average report with incomplete details.	Poor or insufficient documentation submitted.